

Exploratory Data Analysis (EDA) Report

1. Overview

This report analyzes the synthetic dataset generated for Urban Nest Furnishings. The dataset consists of product information and user interaction behavior, used for building a personalized recommender system.

2. Dataset Summary

Products Dataset

- Total Products: 500+
- Attributes: category, material, color, style, price, rating

User Behavior Dataset

- Total Users: 2000+
 - Interactions: 15,000+
 - Event Types: view, cart, wishlist, purchase
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3. Product Analysis

Category Distribution

- Products are distributed across categories like Sofa, Chair, Table, Bed, and Wardrobe
- Distribution is roughly balanced to avoid bias in recommendations

Insight:

Balanced categories ensure the model does not overfit to a single product type.

Price Distribution

- Prices vary based on category:
 - Sofa and Bed → higher range
 - Chair → lower range
- Distribution follows realistic retail pricing patterns

Insight:

Price variation enables meaningful Monetary calculation in RFM.

Product Attributes

- Materials: Wood, Metal, Glass, Fabric, Leather
- Styles: Modern, Vintage, Minimalist, etc.

Insight:

These attributes enable future feature engineering and personalization.

4. User Behavior Analysis

Event Type Distribution

- View events dominate
- Cart and wishlist are moderate
- Purchase events are الأقل but present for every user

Insight:

This reflects realistic user behavior where most interactions are browsing.

User Activity

- Each user has multiple interactions
- Every user has at least one purchase

Insight:

Ensures valid RFM calculation and avoids cold-start issues in Week 1.

5. RFM Analysis (User Features)

Recency

- Measures days since last purchase
- Lower values → more active users

Frequency

- Number of purchases per user
- Higher frequency → loyal customers

Monetary

- Total spend per user
- Higher value → high-value customers

Insight:

RFM segmentation allows grouping users into:

- High-value customers
 - Frequent buyers
 - Inactive users
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6. Data Quality Checks

- No missing values in critical fields
 - Consistent product categories and attributes
 - Valid timestamps within last 1 year
 - Proper linkage between user and product datasets
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7. Key Insights

- User behavior is heavily skewed toward browsing
 - High-value products (sofas, beds) dominate revenue contribution
 - RFM features successfully differentiate user segments
 - Dataset is structured and clean for downstream modeling
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8. Conclusion

The dataset is:

- Complete (meets required scale)
- Clean (no inconsistencies)
- Structured (ready for feature engineering and modeling)

This forms a strong foundation for building recommender systems in subsequent stages.